

Core CS - CN

→ Top 10:



Computer Networks — Top 10 Interview Questions

1. What is the OSI model? Explain its 7 layers.

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes network communication into 7 layers: **Physical** (raw bit transmission), **Data Link** (framing, MAC addressing, error detection), **Network** (IP addressing, routing), **Transport** (end-to-end delivery, TCP/UDP), **Session** (managing sessions between applications), **Presentation** (data translation, encryption, compression), and **Application** (user-facing protocols like HTTP, FTP, DNS). Each layer serves the layer above it and is served by the layer below it.

2. What is the difference between TCP and UDP?

TCP (Transmission Control Protocol) is connection-oriented — it establishes a connection via a 3-way handshake, guarantees delivery, maintains order, and performs error checking and flow control. It's used where reliability matters (HTTP, FTP, email). **UDP** (User Datagram Protocol) is connectionless — it sends datagrams without establishing a connection, offers no delivery guarantee or ordering, but is faster and has lower overhead. It's used where speed is preferred over reliability (DNS, video streaming, VoIP, online gaming).

3. What is the 3-way handshake in TCP?

It's the process TCP uses to establish a reliable connection between a client and server:

- **SYN** — Client sends a SYN packet to the server requesting a connection.
- **SYN-ACK** — Server responds with SYN-ACK, acknowledging the request and sending its own SYN.
- **ACK** — Client sends an ACK to confirm the connection is established.

After this, data transfer begins. Connection termination uses a 4-way FIN handshake.

4. What is the difference between IP address and MAC address?

A **MAC address** (Media Access Control) is a hardware address burned into the NIC — it's 48-bit, operates at the Data Link layer (Layer 2), and is used for communication within a local network. An **IP address** is a logical address assigned to a device — it's 32-bit (IPv4) or 128-bit (IPv6), operates at the Network layer (Layer 3), and is used for routing across networks. MAC addresses don't change; IP addresses can be dynamic (assigned via DHCP).

5. What is subnetting and why is it used?

Subnetting is the process of dividing a large IP network into smaller sub-networks (subnets). It's done by borrowing bits from the host portion of an IP address to create a subnet mask. Benefits include: efficient IP address utilization, reduced broadcast traffic, improved network security and management, and better routing efficiency. For example, a /24 network (255.255.255.0) gives 256 addresses; splitting it into two /25 subnets gives two groups of 128.

6. What is the difference between a hub, switch, and router?

A **hub** operates at Layer 1 — it broadcasts data to all ports regardless of destination, causing collisions and inefficiency. A **switch** operates at Layer 2 — it uses MAC address tables to forward frames only to the intended port, reducing collisions. A **router** operates at Layer 3 — it uses IP addresses to route packets between different networks, enabling inter-network communication. Switches are used within a LAN; routers connect LANs to WANs or the internet.

7. What is DNS and how does it work?

DNS (Domain Name System) translates human-readable domain names (e.g., `google.com`) into IP addresses. When a user types a URL: the browser checks its local cache → queries the OS resolver → asks the configured DNS resolver (usually ISP's) → the resolver queries root nameservers → then TLD nameservers (e.g., `.com`) → then the authoritative nameserver for the domain, which returns the IP. This process is called **recursive resolution**. The result is cached using TTL values to reduce future lookups.

8. What is NAT (Network Address Translation)?

NAT is a technique where a router modifies the IP address information in packet headers while in transit. It allows multiple devices on a private network (using private IPs like 192.168.x.x) to share a single public IP address. **SNAT** (Source NAT) changes the source IP; **DNAT** (Destination NAT) changes the destination IP. NAT conserves IPv4 address space and adds a layer of security by hiding internal IPs from the outside world.

9. What is the difference between HTTP and HTTPS?

HTTP (HyperText Transfer Protocol) is a stateless application-layer protocol used for transferring web data — it sends data in plaintext, making it vulnerable to interception. **HTTPS** is HTTP secured with **TLS/SSL** — it encrypts data in transit, authenticates the server via digital certificates, and ensures data integrity. HTTPS uses port 443 (HTTP uses port 80). The TLS handshake involves certificate verification, key exchange, and establishing a symmetric session key for encryption.

10. What is the difference between congestion control and flow control in TCP?

Flow control prevents the sender from overwhelming the receiver — TCP uses a **receive window** (advertised by the receiver) to indicate how much buffer space is available. The sender limits its transmission accordingly. **Congestion control** prevents the sender from overwhelming the network itself — TCP uses algorithms like **Slow Start**, **Congestion Avoidance**, **Fast Retransmit**, and **Fast Recovery** to dynamically adjust the sending rate based on network conditions (e.g., packet loss signals congestion). Both work together via the concept of an **effective window** = $\min(\text{receive window}, \text{congestion window})$.

→ **Top 25:**



Computer Networks — Top 25 Interview Questions

1. What is the OSI model? Name all 7 layers.

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes how different network protocols interact. The 7 layers, from bottom to top: **Physical** → **Data Link** → **Network** → **Transport** → **Session** → **Presentation** → **Application**. Each layer has a specific role, e.g., Physical handles raw bit transmission, Transport handles end-to-end delivery

(TCP/UDP), and Application provides network services to end-user applications.

2. What is the difference between TCP and UDP?

TCP (Transmission Control Protocol) is connection-oriented, reliable, ordered, and uses error-checking with acknowledgements — used in HTTP, FTP, email. **UDP** (User Datagram Protocol) is connectionless, faster, no guarantee of delivery or order — used in DNS, video streaming, gaming. TCP has higher overhead; UDP is lightweight.

3. What happens when you type a URL in a browser?

1. DNS resolution — domain name is resolved to an IP address.
 2. TCP connection is established (3-way handshake).
 3. HTTP/HTTPS request is sent to the server.
 4. Server processes and sends back a response.
 5. Browser renders the HTML/CSS/JS.
 6. TCP connection is closed (4-way handshake or kept alive).
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4. What is the 3-way handshake in TCP?

It establishes a reliable TCP connection before data transfer:

- **SYN** — client sends a synchronize packet to the server.
- **SYN-ACK** — server acknowledges and sends its own SYN.
- **ACK** — client acknowledges the server's SYN.

After this, the connection is established and data transfer begins.

5. What is DNS and how does it work?

DNS (Domain Name System) translates human-readable domain names (e.g., `google.com`) into IP addresses. Process: browser checks local cache → OS cache → Recursive Resolver → Root Name Server → TLD Name Server → Authoritative Name Server → returns IP. The result is cached at each level using TTL values.

6. What is the difference between HTTP and HTTPS?

HTTP is stateless and transmits data in plaintext. **HTTPS** uses TLS/SSL to encrypt the communication, ensuring confidentiality, data integrity, and server authentication. HTTPS uses port 443; HTTP uses port 80. In HTTPS, a TLS handshake occurs after the TCP handshake to exchange certificates and establish an encrypted session.

7. What is an IP address? Difference between IPv4 and IPv6?

An IP address uniquely identifies a device on a network. **IPv4** is 32-bit (e.g., `192.168.1.1`), supports ~4.3 billion addresses. **IPv6** is 128-bit (e.g., `2001:0db8::1`), supports ~340 undecillion addresses. IPv6 also has built-in IPsec support, eliminates the need for NAT, and uses hexadecimal notation.

8. What is subnetting?

Subnetting divides a large network into smaller sub-networks (subnets) to improve efficiency and security. It uses a **subnet mask** to determine the network and host portions of an IP address. For example, `192.168.1.0/24` means the first 24 bits are the network part, leaving 8 bits for hosts (256 addresses, 254 usable).

9. What is NAT (Network Address Translation)?

NAT allows multiple devices on a private network to share a single public IP address. The router replaces private IPs with its public IP when sending packets out, and maintains a translation table to map responses back to the correct internal device. This conserves IPv4 address space and adds a layer of security.

10. What is the difference between a Hub, Switch, and Router?

- **Hub** — Layer 1, broadcasts data to all ports, no intelligence.
 - **Switch** — Layer 2, uses MAC addresses to forward data only to the intended port, more efficient.
 - **Router** — Layer 3, routes packets between different networks using IP addresses, handles WAN/LAN separation.
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11. What is ARP (Address Resolution Protocol)?

ARP resolves an IP address to a MAC address within a local network. When a device wants to communicate with another device on the same subnet, it broadcasts an ARP request ("Who has IP X?"). The device with that IP replies

with its MAC address. The result is stored in the ARP cache to avoid repeated lookups.

12. What is a MAC address?

A MAC (Media Access Control) address is a unique 48-bit hardware identifier assigned to a network interface card (NIC). It is used for communication within a local network (Layer 2). Unlike IP addresses, MAC addresses are burned into hardware and do not change, though they can be spoofed in software.

13. What is DHCP and how does it work?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration (subnet mask, gateway, DNS) to devices. Process: **DORA** — **Discover** (client broadcasts) → **Offer** (server responds with IP) → **Request** (client accepts) → **Acknowledge** (server confirms). Leases are time-limited and can be renewed.

14. What is the difference between a private and public IP address?

Public IPs are globally unique and routable on the internet, assigned by ISPs. **Private IPs** are used within local networks and are not routable on the internet — ranges: `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`. Devices with private IPs access the internet via NAT.

15. What is a socket in networking?

A socket is an endpoint for communication, defined by an **IP address + port number**. It enables processes on different machines to communicate over a network. Types: **Stream sockets** (TCP, reliable) and **Datagram sockets** (UDP, connectionless). A socket pair (client IP:port ↔ server IP:port) uniquely identifies a connection.

16. What are well-known port numbers?

Ports 0–1023 are reserved for well-known services:

- **80** — HTTP
- **443** — HTTPS
- **21** — FTP
- **22** — SSH
- **25** — SMTP

- **53** — DNS
 - **110** — POP3
 - **143** — IMAP
 - **3306** — MySQL
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17. What is ICMP? How is it used?

ICMP (Internet Control Message Protocol) is used for diagnostic and error-reporting purposes, not for transferring data. It operates at Layer 3. Common uses: **ping** (Echo Request/Reply to test reachability) and **traceroute** (uses TTL expiry to map the path packets take). ICMP doesn't use ports; it has message types and codes instead.

18. What is the difference between connection-oriented and connectionless communication?

Connection-oriented (e.g., TCP) establishes a dedicated connection before data transfer, ensures ordered and reliable delivery, and formally closes the connection afterward. **Connectionless** (e.g., UDP) sends packets independently with no setup or teardown — faster but unreliable. Use connection-oriented for critical data, connectionless for real-time or high-throughput use cases.

19. What is a firewall?

A firewall is a security system that monitors and controls incoming/outgoing network traffic based on predefined rules. Types: **Packet filtering** (inspects headers), **Stateful inspection** (tracks connection state), **Application-layer firewall** (inspects payload, e.g., WAF). Firewalls can be hardware-based, software-based, or cloud-based.

20. What is the difference between LAN, MAN, and WAN?

- **LAN** (Local Area Network) — small geographic area, e.g., office or home, high speed, privately owned.
 - **MAN** (Metropolitan Area Network) — covers a city or campus, e.g., city-wide Wi-Fi.
 - **WAN** (Wide Area Network) — spans large geographic areas, e.g., the internet. Typically uses leased lines or ISP infrastructure.
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21. What is flow control and congestion control in TCP?

Flow control prevents the sender from overwhelming the receiver — implemented via the **sliding window** mechanism where the receiver advertises its buffer size. **Congestion control** prevents the sender from overwhelming the network — TCP uses algorithms like **Slow Start**, **Congestion Avoidance**, **Fast Retransmit**, and **Fast Recovery** to adaptively reduce transmission rate when packet loss is detected.

22. What is a VPN?

A VPN (Virtual Private Network) creates an encrypted tunnel between a client and a server over the public internet, making traffic private and secure. It masks the user's real IP, bypasses geo-restrictions, and allows remote workers to securely access internal networks. Common protocols: OpenVPN, WireGuard, IPSec, L2TP.

23. What is CDN (Content Delivery Network)?

A CDN is a distributed network of servers (edge servers) placed geographically close to users to deliver content faster. When a user requests a resource (image, video, JS), it's served from the nearest edge server rather than the origin server. This reduces latency, improves availability, and lowers load on the origin. Examples: Cloudflare, Akamai, AWS CloudFront.

24. What is the difference between HTTP/1.1, HTTP/2, and HTTP/3?

- **HTTP/1.1** — one request per TCP connection (with keep-alive), suffers from head-of-line blocking.
 - **HTTP/2** — multiplexes multiple requests over a single TCP connection, uses binary framing and header compression (HPACK).
 - **HTTP/3** — built on **QUIC** (over UDP instead of TCP), eliminates TCP head-of-line blocking, faster handshakes, better for lossy networks.
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25. What is SSL/TLS? How does the TLS handshake work?

TLS (Transport Layer Security) encrypts communication between client and server. The handshake:

1. **ClientHello** — client sends supported TLS versions and cipher suites.
2. **ServerHello** — server picks cipher suite, sends its certificate.

3. **Key Exchange** — client verifies the certificate, they exchange keys (e.g., via ECDHE).
4. **Finished** — both sides confirm with a MAC; encrypted session begins.

TLS 1.3 simplifies this to a 1-RTT handshake for faster connections.

→ **Top 50:**

→ **Top 50: Computer Networks Interview Questions**

Q.1 What is a Computer Network?

A computer network is a collection of interconnected devices (nodes) that communicate and share resources using communication channels. Networks can be wired or wireless and follow protocols like TCP/IP for data exchange.

Q.2 What are the types of computer networks?

- **LAN (Local Area Network):** Covers a small area like a building or campus.
 - **WAN (Wide Area Network):** Spans large geographical areas (e.g., the internet).
 - **MAN (Metropolitan Area Network):** Covers a city or large campus.
 - **PAN (Personal Area Network):** Short-range network around an individual (e.g., Bluetooth).
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Q.3 What is the OSI model? Name all 7 layers.

The OSI (Open Systems Interconnection) model is a conceptual framework for how network protocols interact. Layers (top to bottom): **Application** → **Presentation** → **Session** → **Transport** → **Network** → **Data Link** → **Physical**.

Q.4 What is the TCP/IP model? How does it differ from OSI?

The TCP/IP model has 4 layers: **Application, Transport, Internet, Network Access**. Unlike OSI's 7 layers, it merges Presentation and Session into Application, and Physical + Data Link into Network Access. TCP/IP is the practical model used in real networks.

Q.5 What is an IP address? Difference between IPv4 and IPv6?

An IP address is a unique numerical label assigned to each device on a network. **IPv4** uses 32-bit addresses (e.g., 192.168.1.1), supporting ~4.3 billion

addresses. **IPv6** uses 128-bit addresses (e.g., 2001:0db8::1), supporting a vastly larger address space to overcome IPv4 exhaustion.

Q.6 What is subnetting?

Subnetting is the process of dividing a large network into smaller, manageable sub-networks (subnets). It improves performance, security, and efficient IP address utilization. A subnet mask (e.g., 255.255.255.0) defines the boundary between the network and host portions of an IP address.

Q.7 What is a MAC address?

A MAC (Media Access Control) address is a hardware identifier assigned to a Network Interface Card (NIC). It is a 48-bit address written in hexadecimal (e.g., AA:BB:CC:DD:EE:FF) and operates at the Data Link layer. Unlike IP addresses, MAC addresses are globally unique and permanent.

Q.8 What is the difference between TCP and UDP?

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable, guaranteed delivery	Unreliable, no guarantee
Speed	Slower	Faster
Use case	HTTP, FTP, Email	DNS, Video streaming, Gaming

Q.9 What is a three-way handshake in TCP?

The TCP three-way handshake establishes a connection:

1. **SYN** – Client sends a synchronization request to the server.
 2. **SYN-ACK** – Server acknowledges and sends its own SYN.
 3. **ACK** – Client acknowledges the server's SYN. Connection is established.
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Q.10 What is DNS? How does it work?

DNS (Domain Name System) translates human-readable domain names (e.g., [google.com](https://www.google.com)) into IP addresses. When a user enters a URL, the DNS resolver queries a hierarchy of DNS servers (Root → TLD → Authoritative) to return the corresponding IP address.

Q.11 What is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration (subnet mask, gateway, DNS) to devices on a network. It eliminates the need for manual IP configuration and operates using a **DORA** process: Discover → Offer → Request → Acknowledge.

Q.12 What is the difference between a Hub, Switch, and Router?

- **Hub:** Broadcasts data to all devices; operates at Physical layer; no intelligence.
 - **Switch:** Sends data only to the intended device using MAC addresses; operates at Data Link layer.
 - **Router:** Routes data between different networks using IP addresses; operates at Network layer.
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Q.13 What is ARP? How does it work?

ARP (Address Resolution Protocol) maps a known IP address to an unknown MAC address within a LAN. A device broadcasts an ARP request asking "Who has IP X?", and the device with that IP replies with its MAC address. The result is cached in the ARP table.

Q.14 What is NAT?

NAT (Network Address Translation) translates private IP addresses to a public IP address (and vice versa) when traffic passes through a router. It conserves public IP addresses and adds a layer of security by hiding internal network structure from the outside world.

Q.15 What is a Firewall?

A firewall is a network security device (hardware or software) that monitors and controls incoming/outgoing traffic based on predefined security rules. It acts as a barrier between trusted internal networks and untrusted external networks (like the internet).

Q.16 What is HTTP vs HTTPS?

- **HTTP (HyperText Transfer Protocol):** Transfers data in plaintext; not secure.
- **HTTPS:** HTTP secured with SSL/TLS encryption, ensuring data confidentiality and integrity. HTTPS uses port 443, while HTTP uses port 80.

Q.17 What is a VPN?

A VPN (Virtual Private Network) creates an encrypted tunnel over a public network, allowing secure communication as if devices are on a private network. It ensures privacy, anonymity, and bypasses geo-restrictions by masking the user's real IP address.

Q.18 What is the difference between a private and public IP address?

- **Public IP:** Globally unique, assigned by ISPs, used for communication over the internet.
 - **Private IP:** Used within internal networks (e.g., 192.168.x.x, 10.x.x.x); not routable on the internet. NAT is used to map private IPs to a public IP.
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Q.19 What is a Proxy Server?

A proxy server acts as an intermediary between a client and the internet. It forwards client requests to the destination server, providing benefits like anonymity, caching (faster responses), content filtering, and access control.

Q.20 What is ICMP? What is it used for?

ICMP (Internet Control Message Protocol) is used for error reporting and diagnostic functions in IP networks. Tools like **ping** and **traceroute** use ICMP. For example, ping sends ICMP Echo Requests and waits for Echo Replies to test connectivity.

Q.21 What is a port number? List some common ports.

A port number identifies a specific process or service on a host. It allows multiple services to run simultaneously on a single IP.

Port	Protocol
80	HTTP
443	HTTPS
22	SSH
21	FTP
25	SMTP
53	DNS
3306	MySQL

Q.22 What is the difference between connection-oriented and connectionless services?

- **Connection-oriented (e.g., TCP):** A dedicated connection is established before data transfer; ensures reliable, ordered delivery.
 - **Connectionless (e.g., UDP):** No prior connection; data is sent independently as packets with no guarantee of delivery or order.
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Q.23 What is flow control and congestion control?

- **Flow control:** Prevents the sender from overwhelming the receiver by regulating data transmission rate (e.g., TCP's sliding window).
 - **Congestion control:** Prevents network overload by managing the overall traffic in the network (e.g., TCP's slow start, congestion avoidance algorithms).
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Q.24 What is the sliding window protocol?

The sliding window protocol is a flow control mechanism in TCP. It allows multiple frames to be in transit simultaneously without waiting for individual acknowledgments. The window size determines how many bytes can be sent before requiring an ACK, improving throughput.

Q.25 What is a socket?

A socket is an endpoint for communication between two devices over a network. It is defined by a combination of an IP address and a port number. Sockets provide the API for network programming and support both TCP (stream sockets) and UDP (datagram sockets).

Q.26 What is BGP?

BGP (Border Gateway Protocol) is the routing protocol of the internet. It is an **exterior gateway protocol** used to exchange routing information between autonomous systems (AS). BGP makes routing decisions based on paths, network policies, and rule sets, making it the backbone of internet routing.

Q.27 What is OSPF?

OSPF (Open Shortest Path First) is an interior gateway routing protocol that uses **Dijkstra's shortest path algorithm** to find the best route. It is a link-state protocol where each router maintains a complete topology map of the network and updates others when changes occur.

Q.28 What is RIP?

RIP (Routing Information Protocol) is a distance-vector routing protocol that uses **hop count** as the metric to determine the best path (maximum 15 hops). It sends routing table updates to neighbors every 30 seconds. RIP is simple but not scalable for large networks.

Q.29 What is the difference between static and dynamic routing?

- **Static routing:** Routes are manually configured by an administrator; no overhead but not scalable.
- **Dynamic routing:** Routers automatically discover and update routes using protocols like OSPF, RIP, or BGP; adapts to network changes but has protocol overhead.

Q.30 What is VLAN?

A VLAN (Virtual LAN) logically segments a physical network into multiple isolated broadcast domains without requiring separate hardware. It improves security, reduces broadcast traffic, and allows grouping of devices regardless of physical location. Configured on managed switches using 802.1Q tagging.

Q.31 What is the difference between half-duplex and full-duplex?

- **Half-duplex:** Communication in both directions, but only one direction at a time (e.g., walkie-talkies).
- **Full-duplex:** Simultaneous two-way communication (e.g., telephone calls, modern Ethernet switches).

Q.32 What is bandwidth vs throughput vs latency?

- **Bandwidth:** Maximum data transfer capacity of a link (theoretical), measured in bps.
- **Throughput:** Actual data transfer rate achieved in practice (always $\leq$ bandwidth).
- **Latency:** Time taken for a packet to travel from source to destination; also called delay.

Q.33 What is a CDN?

A CDN (Content Delivery Network) is a distributed network of servers that caches and delivers content (web pages, videos, images) from locations geographically closer to users. This reduces latency, improves load times, and decreases the load on the origin server.

Q.34 What is SSL/TLS?

SSL (Secure Sockets Layer) and its successor TLS (Transport Layer Security) are cryptographic protocols that provide secure communication over a network. They use a combination of asymmetric encryption (for key exchange) and symmetric encryption (for data transfer), ensuring **confidentiality, integrity, and authentication**.

Q.35 What is a digital certificate and how does it work?

A digital certificate is an electronic document issued by a Certificate Authority (CA) that binds a public key to an entity's identity. It is used in HTTPS to verify that a server is legitimate. The browser checks the certificate's signature against trusted CAs to establish a secure connection.

Q.36 What is the difference between symmetric and asymmetric encryption?

- **Symmetric encryption:** Same key for encryption and decryption (e.g., AES). Fast but key distribution is a challenge.
 - **Asymmetric encryption:** Uses a public key to encrypt and a private key to decrypt (e.g., RSA). Slower but solves the key distribution problem.
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Q.37 What is a DDoS attack?

A DDoS (Distributed Denial of Service) attack floods a server or network with massive traffic from multiple compromised machines (botnet), overwhelming it and making it unavailable to legitimate users. Mitigations include rate limiting, firewalls, and CDN-based traffic scrubbing.

Q.38 What is a Man-in-the-Middle (MITM) attack?

In a MITM attack, an attacker secretly intercepts and potentially alters communication between two parties who believe they are directly communicating. It can be used to eavesdrop, steal credentials, or inject malicious data. HTTPS and certificate pinning are common defenses.

Q.39 What is the difference between FTP and SFTP?

- **FTP (File Transfer Protocol):** Transfers files over port 21 in plaintext; no encryption; insecure.
 - **SFTP (SSH File Transfer Protocol):** Transfers files over SSH (port 22) with full encryption; secure alternative to FTP.
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Q.40 What is SMTP, POP3, and IMAP?

- **SMTP (Simple Mail Transfer Protocol):** Used to **send** emails (port 25/587).
 - **POP3 (Post Office Protocol 3):** Downloads emails from server to client and deletes from server (port 110).
 - **IMAP (Internet Message Access Protocol):** Syncs emails between server and multiple clients without deleting from server (port 143).
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Q.41 What is a broadcast, unicast, and multicast?

- **Unicast:** One-to-one communication (single sender, single receiver).
 - **Broadcast:** One-to-all communication within a network segment (e.g., ARP requests).
 - **Multicast:** One-to-many communication to a specific group of interested receivers (e.g., video conferencing).
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Q.42 What is CIDR notation?

CIDR (Classless Inter-Domain Routing) notation represents an IP address and its subnet mask in a compact form (e.g., 192.168.1.0/24). The number after the slash indicates how many bits are used for the network portion. It allows flexible allocation of IP address blocks and replaced the rigid class-based system.

Q.43 What is a loopback address?

The loopback address (127.0.0.1 in IPv4, ::1 in IPv6) is a special address used to test the network stack on the local machine. Traffic sent to 127.0.0.1 never leaves the device; it is used to verify that TCP/IP is installed and functioning correctly.

Q.44 What is the difference between TCP's SYN flood and normal connection?

In a normal TCP connection, a three-way handshake (SYN → SYN-ACK → ACK) completes successfully. In a **SYN flood attack**, the attacker sends many SYN

packets but never completes the handshake, leaving the server with many half-open connections that exhaust resources. SYN cookies are a common mitigation.

Q.45 What is Quality of Service (QoS)?

QoS is a set of techniques used to manage network resources by prioritizing certain types of traffic (e.g., VoIP, video conferencing) over others (e.g., file downloads). It ensures that latency-sensitive applications receive sufficient bandwidth and low delay even under network congestion.

Q.46 What is a network topology? Name the types.

Network topology defines the physical or logical arrangement of devices in a network.

- **Bus:** All devices share a single cable.
 - **Star:** All devices connect to a central hub/switch.
 - **Ring:** Devices connected in a circular loop.
 - **Mesh:** Every device connects to every other device.
 - **Tree/Hybrid:** Combination of topologies.
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Q.47 What is the difference between a gateway and a router?

A **router** connects multiple networks and forwards packets based on IP addresses. A **gateway** is a broader concept — it acts as an entry/exit point between networks that may use different protocols, performing protocol translation if needed. Every router can be a gateway, but not every gateway is a router.

Q.48 What is traceroute and how does it work?

Traceroute is a diagnostic tool that shows the path packets take from source to destination, listing each hop (router) along the way. It works by sending packets with incrementally increasing TTL (Time To Live) values. Each router decrements the TTL by 1; when it reaches 0, an ICMP "Time Exceeded" message is sent back, revealing that router's address.

Q.49 What is the difference between circuit switching and packet switching?

- **Circuit switching:** A dedicated communication path is established for the entire session (e.g., traditional telephone networks). Guaranteed bandwidth

but inefficient if idle.

- **Packet switching:** Data is broken into packets sent independently across the network (e.g., the internet). Efficient and scalable but no guaranteed delivery order.

Q.50 What is SDN (Software Defined Networking)?

SDN is a networking approach that separates the **control plane** (decision-making) from the **data plane** (packet forwarding). A centralized software controller manages the network behavior dynamically, making networks more programmable, flexible, and easier to manage compared to traditional hardware-based routing and switching.